

Data Warehouse: A centralized database that integrates historical and current data from different systems for managerial analysis.

Data Mining: Techniques used to discover patterns from large volumes of data, texts, or websites.

Business Intelligence: A set of tools and systems that allow companies to collect, store, analyze, and access data to make informed and strategic decisions.

VPN (Virtual Private Network): Technology that creates a secure and encrypted connection over a public network such as the Internet, allowing users to access network resources remotely and securely.

Botnet: A network of computers infected with malicious software that are remotely controlled by an attacker, typically used to perform malicious activities such as denial-of-service attacks or massive spam distribution.

Cracker: An individual who illegally accesses computer systems with the intent to cause harm or steal information, differing from ethical hackers who seek to improve security.

ERM (Enterprise Risk Management): A process through which an organization identifies, evaluates, and manages risks that could affect the achievement of its objectives, integrating risk management at all levels of the company.

Customer Churn Rate: The percentage of customers who stop using a company's products or services during a specific period—this is a key indicator for evaluating customer retention.

Banner: A graphic advertisement displayed on a webpage, usually in the form of an image or animation, designed to attract the user's attention and direct them to another website.

Digital Goods: Products that are stored, delivered, and used in digital format, such as software, music, videos, e-books, and other digital content.

1. What is normalization? How does it relate to the characteristics of a well-designed relational database?

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves dividing large tables into smaller ones and defining relationships between them. A well-designed relational database uses normalization to ensure that each piece of data is stored only once, making it easier to maintain and reducing the likelihood of inconsistencies.

2. Define the following: WAN, MAN, 3G, modem, protocol, optical network, bandwidth, and Internet2

- **WAN (Wide Area Network):** A network that covers a large geographic area, such as a country or continent. The Internet is the most well-known example of a WAN.
- **MAN (Metropolitan Area Network):** A network that covers a city or metropolitan area, connecting multiple local area networks.
- **3G:** Third generation of mobile technology that allows high-speed data transmission, enabling services such as video calls and mobile internet access.
- **Modem:** A device that converts digital signals into analog signals and vice versa, enabling data transmission over telephone lines or other media.
- **Protocol:** A set of rules that govern communication between devices on a network, ensuring that data is transmitted correctly.
- **Optical Network:** A communication network that uses fiber optics to transmit data at high speeds over long distances with minimal signal loss.
- **Bandwidth:** The maximum data transmission capacity of a network or connection over a given period, usually measured in bits per second (bps).
- **Internet2:** An advanced research and development network that provides high-speed networking capabilities to academic and research institutions.

3. Why is it so difficult to protect the Internet and Wi-Fi networks?

Protecting the Internet and Wi-Fi networks is complex due to their open and decentralized nature. Easy access, the wide variety of connected devices, and the constant evolution of threats make maintaining security a challenge. Additionally, many users do not implement adequate security measures, increasing network vulnerability.

4. How do enterprise systems add value to a business?

Enterprise systems integrate and automate key business processes such as finance, human resources, and production. This improves operational efficiency, provides real-time information for decision-making, and facilitates coordination among different departments—resulting in a competitive advantage for the company.

5. Mention and describe four business trends and three technology trends that are shaping today's e-commerce.

Business Trends:

1. **Globalization:** Companies expand operations beyond national borders, reaching new markets and customers.
2. **Personalization:** Offering products and services tailored to individual customer preferences to enhance their experience.
3. **Sharing Economy:** Business models that allow people to share resources, such as transportation or lodging, through digital platforms.
4. **Digital Marketing:** Using digital channels to promote products and services, reaching specific audiences more effectively.